INCH-POUND
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DETAIL SPECIFICATION

DOLLY, TRAILER CONVERTORS, M197A1, M198A1 AND M354

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers M197A1, M198A1 and M354 tactical dollies for use in converting various semitrailers to full trailers.

1.2 Classification. Dollies will be of the following models as specified (see 6.2 and 6.4). Dollies will also be referred to herein as "vehicles."

Model	Description
M197A1	Dolly, trailer convertor, 6 ton, 2 wheel
M198A1	Dolly, trailer convertor, 8 ton, 2 wheel
M354	Dolly, trailer convertor, 15 ton, 4 wheel

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2330

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50271 - Plates, Identification, Instruction and Marking, Blank.

DEPARTMENT OF DEFENSE

MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant (CARC).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicle Standard (FMVSS)

FMVSS No. 105 - Hydraulic Brake System.

(Application for copies should be addressed to the U.S. Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DRAWINGS

ARMY

8736120 - Dolly, Trailer Converter, 6 Ton, 2 Wheel, M197A1.
8736124 - Dolly, Trailer Converter, 8 Ton, 2 Wheel, M198A1.
8736308 - Dolly, Trailer Converter, 15 Ton, 4 Wheel, M354.

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(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

GENERAL MOTORS CORPORATION

GM 9540P - Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, c/o Global Engineering, 15 Inverness Way, Englewood, CO 80112.)

THE TIRE AND RIM ASSOCIATION, INC.

Year Book.

(Application for copies may be obtained from The Tire and Rim Association, Inc., 175 Montrose West Ave., Suite 150, Copley, OH 44321.)

EUROPEAN TYRE AND RIM TECHNICAL ORGANIZATION (ETRTO)

Standards Manual.

(Application for copies may be obtained from the European Tyre and Rim Technical Organization, 32 Avenue, Brugmann, 1060 Brussels, Belgium.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for dollies. The materials shall be capable of meeting all the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent practicable (see 4.5.1).

3.2.1 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings (see 4.5.1 and 6.2).

3.3 Construction. The vehicle shall be constructed in accordance with the applicable drawings specified in table I and as supplemented herein. The vehicle shall be complete with all specified items installed, including reflectors, payload tiedowns, lashing hooks, lifting devices, and tools (see 4.5.2).

TABLE I. Drawings.

Model	Drawing
M197A1	8736120
M198A1	8736124
M354	8736308

3.3.1 Seals. When fording or operating in mud, sand or snow, seals shall prevent entrance of foreign matter into the bearings. Bearing seals shall restrict the leaking of lubricants from bearing (see 4.5.3).

3.3.2 Electrical circuits. Electrical circuits shall maintain continuity from end to end, without evidence of internal or external shorts, during all vehicle operating conditions (see 4.5.2 and 4.5.4).

3.3.3 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction under all vehicle operating conditions (see 4.5.2 and 4.5.5).

3.3.4 Adjustment mechanisms. All adjustment mechanisms shall function properly, and maintain adjustment within required settings during all vehicle operating conditions (see 4.5.2 and 4.5.6).

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3.3.5 Welding and rivets. Welding and rivets shall meet the quality requirements specified on the applicable drawing (see 4.5.2).

3.3.5.1 Welding repairs. Welding repairs of any kind shall be made only when, and to the extent, specifically authorized by the procuring activity (see 4.5.2 and 6.2).

3.3.6 Tires and tubes.

3.3.6.1 Tires. Tires shall be cross-country type as specified for tactical vehicles. Tires shall be in accordance with the Tire and Rim Association or to the European Tyre and Rim Technical Organization recommendations. Size of tires shall be as specified in the applicable vehicle characteristic data sheets (see 4.5.1, 4.5.2 and 6.4).

3.3.6.2 Inner tubes. When tube type tires are specified, inner tubes shall be of heavy duty type and shall be of proper size for tires furnished. Tube type radial tires shall have radial type inner tubes. Tire flaps shall be provided for tube type tires in accordance with the Tire and Rim Association recommendations.

3.3.7 Intervehicular connections. Safety chains, cables, and hoses shall be of sufficient length and routed correctly to function properly under any vehicular orientation (see 4.5.2).

3.3.7.1 Drawbar assembly. Drawbar shall raise and lower correctly to allow vehicular connection (see 4.5.2).

3.3.8 Suspension. Suspension clearances shall prevent interference between wheels and other portions of the vehicle when vehicle is in use under any condition of its intended purpose (see 4.5.2 and 4.5.7).

3.3.9 Axle assembly. The axle assembly shall be properly installed and shall have a capacity as specified in table I for the applicable vehicle (see 4.5.2).

3.3.10 Fifth wheel. The fifth wheel shall accommodate the applicable semitrailer kingpin and meet the performance requirements of the 6000 mile operational test (see 4.5.2 and 4.5.7).

3.3.11 Lunette. Lunette shall accommodate the applicable towing vehicle attachment device at maximum gross vehicle weight (GVW) of the trailer (see 4.5.2).

3.4 Performance. The vehicle shall meet the performance requirements of this specification when fully equipped, loaded with rated payload, serviced with standard products (see 6.5) and towed by appropriate military towing vehicle.

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3.4.1 Reliability. Vehicles with rated payload, when operated in accordance with 4.5.7, shall possess not less than 60 percent probability of completing not less than 6000 miles without replacement or overhaul of the principal components and/or subassemblies.

3.4.2 Maintainability. The total scheduled maintenance, excluding driver/crew checks and services, shall be between 4 and 8 manhours. The total unscheduled maintenance shall be between 0 and 12 manhours during the first 6000 miles of operation (see 4.5.7). This equates to a Maintenance Ratio (MR) of 0.013 to 0.067 at 20 miles per hours (mph) and a MR of 0.02 to 0.10 at 30 mph. The scheduled maintenance intervals shall be 3 months or 3000 miles of operation, whichever comes first. Maximum time to repair (only to include diagnosis, repair and verification) shall be as follows, using personnel normally employed:

a. Organization maintenance shall not exceed 3 manhours 95% of the time.
b. Direct support maintenance shall not exceed 10 manhours 95% of the time.
c. General support maintenance - not required.

3.4.3 Environmental. The vehicle shall operate as specified herein, under extreme conditions of weather, in ambient air temperatures ranging from -50 degrees Fahrenheit (°F) to +125°F (see 4.5.8).

3.4.4 Service products. Servicing and lubrication shall be performed using standard service products (see 4.5.2 and 6.5).

3.4.5 Highway and cross-country operations.

3.4.5.1 Highway operation. When loaded and towed at specified speeds, in a straight line, on a relatively level, smooth paved surface, the path of the towed vehicle shall not deviate more than 3 inches to either side of the path of the towing vehicle (see 4.5.9).

3.4.5.2 Cross-country operation. The vehicle shall be towable at specified load and speed over unimproved roads, or gently rolling terrain. When so operated in a straight line, the vehicle shall track the towing vehicle without damage to itself or the towing vehicle. Stability shall be maintained (see 4.5.10).

3.4.6 Gradeability. When towed over a dry, hard-surfaced road, free of loose material, each vehicle shall exhibit the stability and gradeability specified.

3.4.6.1 Longitudinal inclines. Each vehicle shall follow the towing vehicle, without weaving to an extent which adversely affects the controllability of the vehicle combination, when ascending or descending longitudinal inclines having a 15% grade (see 4.5.11).

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3.4.6.2 Side slopes. Each vehicle shall follow the towing vehicle without slipping or upsetting when operating, either left or right, on 20% side slopes (see 4.5.12).

3.4.7 Turning ability. When coupled to the towing vehicle, operating in its minimum turning circle, each vehicle shall follow without cramping or side-slipping and without damage to or interference between the towed vehicle and the towing vehicle (see 4.5.13).

3.4.8 Braking ability.

3.4.8.1 Service brakes. The combined service brakes of the vehicle and towing vehicle shall control, decelerate, and stop the vehicle combination within 30 feet from a speed of 20 mph on dry, hard, level, smooth surfaced roads. Hydraulic brakes shall conform to FMVSS No. 105 (see 4.5.2 and 4.5.14).

3.4.8.2 Automatic actuation. The automatic breakaway device shall apply and maintain application of the vehicle brakes for not less than 15 minutes, upon breakaway from the towing vehicle (see 4.5.2 and 4.5.15).

3.4.9 Fording. Each vehicle shall ford hard-bottomed, salt or fresh water crossings of depths great enough to completely submerge the chassis to top of fifth wheel and shall withstand such submergence for periods up to 15 minutes duration without damage to the vehicle or its components (see 4.5.16).

3.4.10 Landing leg. Unless otherwise specified (see 6.2), the vehicle shall have a landing leg. The landing leg shall exhibit capacity and adjustment to raise, support and hold its portion of the fully loaded vehicle and shall withstand, without damage, the strains imposed when coupling or uncoupling the towing vehicle. The landing leg shall extend and retract; and when retracted, shall provide ground clearance for cross-country operation (see 4.5.2 and 4.5.17).

3.5 Surface treatment. Unless otherwise specified (see 6.2), exterior surfaces of the vehicle shall be cleaned, treated, primed, and painted in accordance with the manufacturer's standard practice. The color of the exterior paint coat shall be as specified in the contract (see 6.2). When specified (see 6.2), Chemical Agent Resistant Coatings (CARC) paint shall be required (see 4.5.2).

3.5.1 Corrosion protection. The dollies shall function as intended and shall show no signs of detrimental corrosion after being subjected to a salt spray solution (see 4.5.8.1).

3.6 Marking. The identification marking and data plates shall be permanent and legible and shall include, as a minimum (see 4.5.2, 6.2 and 6.3):

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a. Part or identification number (PIN).
b. Model number.
c. Manufacturer's name or trademark.
d. Date of manufacture.
e. Contract or order number.

3.6.1 Identification plates. Identification plates shall meet the requirements of A-A-50271 (see 4.5.2).

3.7 Safety. All exposed parts which are energized electrically shall be located, insulated, fully enclosed, or guarded so as to prevent hazards to operating or maintenance personnel, or interference with equipment functioning. All moving parts which are of such nature, or so located, as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair operating functions (see 4.5.18).

3.8 Servicing. The vehicle shall be adjusted for immediate operation and shall be serviced for climatic conditions at the delivery point. Servicing shall be compatible with processing for shipment as specified by the procuring activity (see 4.5.2 and 5.1).

3.9 Workmanship. Workmanship shall be of such quality as to assure that the vehicle and its components are free of defects that compromise, limit or reduce the capability of the vehicle in the performance of its intended use. In addition, the vehicle shall be free of defects which affect its appearance (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (CI) (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on the first production vehicle and on two additional vehicles randomly selected from the first month's production or from the first 20 produced as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.3 through 4.5.18 (see table III).

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TABLE II. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Construction, improper dimensions affecting performance (see 3.3).	SIE 1/
102	Electrical system, malfunctioning, inoperative, damaged (see 3.3.2).	Visual/ functional
103	Controls, adjustment mechanism; malfunctioning, inoperative, damaged (see 3.3.3 and 3.3.4).	Visual/ functional
104	Frame; structural welding and riveting defects (see 3.3.5).	Visual
105	Wheels, tires and tubes; not as specified, tire or tube damage (see 3.3.6).	Visual/SIE
106	Safety chains, intervehicular cable and hose; improper lengths, damaged, malfunctioning, missing (see 3.3.7).	Visual/ dimensional
107	Drawbar assembly; raising and lowering action improper (see 3.3.7.1).	Visual/ functional
108	Suspension system components; improper clearances, leaking, noisy (see 3.3.8).	Visual/ functional
109	Axle; improper assembly or installation (see 3.3.9).	Visual
110	Fifth wheel; malfunctioning, improper assembly or installation (see 3.3.10).	Visual/ functional
111	Lunette eye; casting defect, improper size (see 3.3.11).	Visual/SIE
112	Brake system; malfunctioning, component damage, nonconformance (see 3.4.8).	Visual/SIE
113	Landing leg; improper extension, retraction, stowage action, locking action (see 3.4.10).	Visual/ functional
Minor:		
201	Construction, improper dimensions not affecting performance (see 3.3).	SIE
202	Reflectors; missing, defective, improper installation (see 3.3).	Visual
203	Tiedowns, lashing hooks, and lifting devices; missing, improperly located (see 3.3).	Visual
204	Tools; missing (see 3.3).	Visual
205	Electrical system components; improper assembly or installation (see 3.3.2).	Visual
206	Wheels, tires and tubes; improper mounting, pressure, or stud torque (see 3.3.6).	Visual/ tactile
207	Drawbar assembly; improper assembly or installation (see 3.3.7.1).	Visual

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TABLE II. Classification of defects - Continued.

Category	Defect	Method of examination
208	Suspension system components, including springs and shock absorbers; improper assembly or installation (see 3.3.8).	Visual
209	Lunette eye; improper assembly or installation (see 3.3.11).	Visual
210	Brake components; improper assembly or installation, clearances inadequate (see 3.4.8).	Visual
211	Landing leg; improper assembly or installation (see 3.4.10).	Visual
212	Paint and protective coating; improper application, coverage, or color (see 3.5).	Visual
213	Marking; missing, incomplete, improper size (see 3.6).	Visual
214	Identification plates and markings; missing, incomplete, improper size, improper installation (see 3.6).	Visual
215	Lubrication fitting; missing, improper installation, defective (see 3.8).	Visual
216	Lubrication; improper application (see 3.8).	Visual
217	Workmanship; affecting appearance (see 3.9).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE III. Classification of inspections.

Title	Requirements	Inspection	First article		Conformance
			Vehicle 1	Vehicles 2 and 3	
Material and construction	3.2 thru 3.3, 3.3.6 and 3.4.4	4.5.1	X		
Defects	3.3, 3.3.2 thru 3.3.11, 3.4.8, 3.4.10, 3.5, 3.6, 3.6.1 and 3.8 thru 3.9	4.5.2			X
Seals	3.3.1	4.5.3	1/	1/	
Electrical circuits	3.3.2	4.5.4	1/	1/	1/
Controls	3.3.3	4.5.5	1/	1/	1/
Adjustment mechanism	3.3.4	4.5.6	1/	1/	1/
Reliability	3.4.1	4.5.7		X	
Maintainability	3.4.2	4.5.7		X	
Environmental	3.4.3	4.5.8	X	X	
Highway operation	3.4.5.1	4.5.9	X	X	X

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TABLE III. Classification of inspections - Continued.

Title	Requirements	Inspection	First article		Conformance
			Vehicle 1	Vehicles 2 and 3	
Cross-country operation	3.4.5.2	4.5.10	X	X	
Longitudinal inclines	3.4.6.1	4.5.11	X	X	
Side slopes	3.4.6.2	4.5.12	X	X	
Turning ability	3.4.7	4.5.13	X	X	
Service brakes	3.4.8.1	4.5.14	X	X	X
Automatic actuation	3.4.8.2	4.5.15	X	X	
Fording	3.4.9	4.5.16	X	X	
Landing legs	3.4.10	4.5.17	X	X	X
Corrosion protection	3.5.1	4.5.8.1	X		
Safety	3.7	4.5.18	1/	1/	

1/ These examinations shall be part of the applicable road test.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 (see table II), followed by 5 miles of operation on relatively hard-surfaced roads, followed by the tests of 4.5.4 through 4.5.6, 4.5.9, 4.5.14 and 4.5.17.

4.3.1 Acceptance tests (100 percent). Unless otherwise specified (see 6.2), each vehicle shall be subject to the conformance inspection.

4.3.2 Sampling. If one hundred percent conformance inspection is waived (see 6.2), sampling inspection shall be conducted. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4.

4.4 Examination. Each vehicle shall be examined for compliance with the requirements specified in table II. Any design or modification of the contractor's standard to comply with specified requirements of the examination shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of applicable requirements listed in table II. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Methods of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records,

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processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3, 3.3.2 through 3.3.11, 3.4.8, 3.4.10, 3.5, 3.5.1, 3.6, 3.6.1 and 3.8 through 3.9, shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile or by measurement with SIE.

4.5.3 Seals. To determine conformance to 3.3.1, after the 6000 mile test and fording operation, all bearings shall be checked for dirt contamination and lubricant leakage.

4.5.4 Electrical circuit. To determine conformance to 3.3.2, current and voltage output levels and continuity of the electrical system shall be measured.

4.5.5 Controls. To determine conformance to 3.3.3, all controls shall be operated and examined for functional requirements.

4.5.6 Adjustment mechanisms. To determine conformance to 3.3.4, all adjustment mechanisms shall be examined for required adjustment and function.

4.5.7 6000 mile operational test. To determine conformance to 3.4.1 and 3.4.2, the vehicles shall be subjected to a 6000 mile operational test as specified in table IV and table V. The vehicle shall complete the operation without replacement or overhaul of the suspension, including springs, axles, and wheels, major brake system components (less brake shoes and common hardware), landing device, lunette, drawbar and frame.

TABLE IV. Vehicle loads and speeds, for 6000 mile test.

Model	Rated payload 1/ (pounds)	Total 2/ weight (pounds)	Towing vehicle	Semitrailer class	Maximum speeds (mph)	
					Highway	Cross-country
M197A1	12 000	14 880	2-1/2 ton	6 ton	50	10
M198A1	16 000	19 450	5 ton	8 ton	50	10
M354	30 000	36 100	10 ton	15 ton	35	10

1/ Payloads indicated are for highway and cross-country operations. If the dolly and the empty semitrailer reach the total combined weight requirement no semitrailer payload is required.

2/ Combined weight of dolly and payload.

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TABLE V. 6000 mile test.

Course	Mileage and speeds 1/
Hard-surfaced, paved	2400 miles at varying speeds up to maximum highway speed with rated payload.
Hilly cross-country (Churchville "C")	600 miles at varying speeds up to maximum cross-country speed with rated payload.
Level cross-country	1440 miles at varying speeds up to maximum cross-country speed with rated payload.
Munson improved gravel roads	1500 miles at varying speeds up to maximum cross-country speed with rated payload.
Belgian block	60 miles at varying speeds up to maximum cross-country speed (40 miles shall be with rated payload and 20 miles shall be with no payload).

4.5.8 Environmental. To determine conformance to 3.4.3, the vehicle, properly serviced and equipped, shall be operated at the temperatures specified. The vehicle shall evidence no damage as a result of such operation.

4.5.8.1 Corrosion protection test. To determine conformance to 3.5.1, the finished dollies shall be subjected to the Accelerated Corrosion Test of GM 9540P, Method B, 100 cycles. Following the test, all moving components of the dollies shall be operated and the dollies shall be examined for detrimental corrosion.

4.5.9 Highway operation. To determine conformance to 3.4.5.1, the vehicle shall be coupled to the towing vehicle. The combination shall be operated on paved highway and shall meet the performance specified.

4.5.10 Cross-country operation. To determine conformance to 3.4.5.2, the vehicle shall be operated over cross-country terrain, as specified, without damage to the vehicle or towing vehicle.

4.5.11 Longitudinal inclines. To determine conformance to 3.4.6.1, the vehicle shall be towed up and down longitudinal inclines at cross-country speed and shall meet specified requirements.

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4.5.12 Side slopes. To determine conformance to 3.4.6.2, the vehicle shall be towed on side slopes at cross-country speed and shall meet specified requirements.

4.5.13 Turning ability. To determine conformance to 3.4.7, the towing vehicle shall be coupled to the vehicle, driven to the towing vehicle's minimum turning circle, to the right and to the left, without interference between the towing and the towed vehicles.

4.5.14 Service brakes. To determine conformance to 3.4.8.1, the vehicle combination shall be operated as specified and observed for stopping distance. Stopping distance shall not exceed that specified. During CI, the vehicle braking system shall be operated to the extent necessary to ensure proper operation and adjustment of service brakes on all wheels.

4.5.15 Automatic actuation brake. To determine conformance to 3.4.8.2, the vehicle and towing vehicle shall be placed on 15% grade and the automatic break-away device actuated. The brakes shall hold the vehicle stationary for time specified, when uncoupled from towing vehicle.

4.5.16 Fording. To determine conformance to 3.4.9, the vehicle shall be towed through water crossing at the depth and for the time period specified.

4.5.17 Landing leg. To determine conformance to 3.4.10, the vehicle shall be coupled and uncoupled from the towing vehicle and the landing leg operated. The landing leg shall perform as specified.

4.5.18 Safety. To determine conformance to 3.7, the vehicle shall be examined to assure that hazards to operating and maintenance personnel have been eliminated or controlled.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

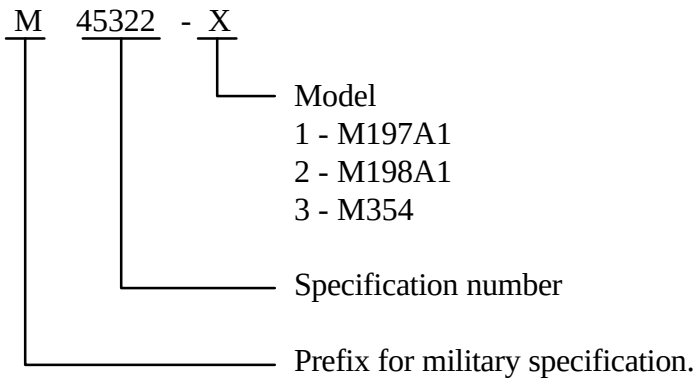
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6.1 Intended use. Dollies covered by this specification are intended for use in converting semitrailers to full trailers for transporting military supplies and equipment, and for mounting special equipment, when required. Due to the extreme battlefield conditions under which they are designed to operate, these dollies are considered military unique.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Model, type and other identifying data of the vehicle (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article is required (see 3.1).
e. If ozone resistant rubber components are required (see 3.2.1).
f. If welding repairs are authorized and extend allowed (see 3.3.5.1).
g. If a landing leg is not required (see 3.4.10).
h. Color of exterior paint coat (see 3.5).
i. If CARC paint is required (see 3.5).
j. PIN and quantity of vehicles required (see 3.6 and 6.3).
k. If 100 percent conformance examination is not required (see 4.3.1).
l. If conformance sampling inspection is to be conducted (see 4.3.2).
m. Packaging requirements (see 5.1).

6.3 Part or identifying number (PIN). The PIN to be used for vehicles acquired to this specification are created as follows:



6.4 Vehicle characteristic data. Vehicle characteristic data for the dollies covered by this specification is provided in the following vehicle characteristic data sheets.

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Model	Data sheet
M197A1	AT-VCS-0214-M197A1
M198A1	AT-VCS-0215-M198A1
M354	AT-VCS-0226-M354

6.5 Recommended service products. Typically, the dolly trailer should be serviced with the following product listed in table VI or equivalent.

TABLE VI. Service products.

Product use	Ambient air temperature	
	-65 to 0°F	-10 to +132°F
Oil for hydraulic brakes	Silicone, all weather automotive	SAE J1783
Oil for general purpose lubrication	General purpose, preservative (water-displacing, low temperature)	MIL-L-3150
Oil for shock absorbers	MIL-H-5606	MIL-H-5606
Grease for sealed bearings	MIL-G-23827	MIL-G-23827
Grease for general chassis lubrication and wheel bearings	MIL-G-10924	MIL-G-10924

6.6 Subject term (key word) listing.

Drawbar
Fifth wheel
Landing leg
Towing

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - MC

Preparing Activity:
Army - AT

(Project 2330-0307)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
	MIL-DTL-45322H	980904

3. DOCUMENT TITLE

DOLLY, TRAILER CONVERTORS, M197A1, M198A1 AND M354

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
	(810) 574-8745 786-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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